PAPER_1544_RYDBERG_E_R_13_6057

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PAPER_1544 — Rydberg Energy $E_R = E_R = D_{phys} + SO_5 - F_{TRZ} \cdot D_{phys} + F_{TRZ}^2 \cdot SSQ = 4 + 10 - 0.4 + 0.0057 = 13.6057$ eV

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Quantum/Atomic
Date: June 18, 2026
Status: CLOSED — EXACT closure

Observation

PAPER_1209EE_S623 (Quantum/Atomic Unified Proof Set) lists **Rydberg Energy E_R** as a closure with the formula:

``
```
Rydberg Energy E_R = E_R = D_phys + SO_5 – F_TRZ·D_phys + F_TRZ²·SSQ = 4 + 10 –
0.4 + 0.0057 = 13.6057 eV
```
``

Status tier: **EXACT**. Units: eV.

UQFF Closed Identity

``
```
E_R = D_phys + SO_5 – F_TRZ·D_phys + F_TRZ²·SSQ = 4 + 10 – 0.4 + 0.0057 =
13.6057 eV EXACT
```
``

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Rydberg Energy E_R is a fundamental quantum/atomic constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to "replace" the experimental value.

Reference

- Source: PAPER_1209EE_S623
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "rydberg_e_r_13_6057"})`
